## ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(c)** 3822 Pa | Area = 3.14 x 0.01 = 0.0314 square m; 120/0.0314 = 3822 Pa. |
| 3 | **(b)** A is likely to be stronger | Sea supplies energy. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 5 | **(c)** 4000 Pa | Inversely proportional. |
| 6 | **(a)** wind and waves are strongest near the eye wall | Danger. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(d)** 60 N upward | Balanced forces. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | The sharp edge has a smaller area, so the pressure is greater. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | Greater pressure at the bottom needs a stronger and thicker wall. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Area = 0.005 square metre (1); P = 50/0.005 = 10000 Pa (1). | 1+1 |
| 12 | They show cloud patterns over a large area (1); Help in predicting path and intensity (1). | 1+1 |
| 13 | Wind lowers pressure above the roof (1); Higher pressure inside pushes it up (1). | 1+1 |
| 14 | P = 450/0.0009 = 500000 Pa (1); Very high; can damage floors (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Area = 0.01 square m; P = 50/0.01 = 5000 Pa (1); Stacked: weight 100 N (1); P = 10000 Pa (1). | 1+1+1 |
| 16 | Pressure gradient is steepest (1); Air rushes toward the centre (1); Rotation increases speed (1). | 1+1+1 |
| 17 | Helium is lighter than air (1); Surrounding air provides upward push (1); Air-filled balloon is not lighter (1). | 1+1+1 |
| | **OR:** Feet: 700/0.04 = 17500 Pa (1); Skis: 700/0.4 = 1750 Pa (1); Skis reduce pressure to one-tenth (1). | 1+1+1 |
| 18 | Falls on approach (1); Lowest at the eye (1); Rises after it passes (1). | 1+1+1 |
| | **OR:** Crops are damaged (1); Soil erosion (1); Early warnings and insurance (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Pressure = force / area (1). (ii) Area = 0.02 square metre, P = 600 / 0.02 = 30000 Pa (1). (iii) P = 600 / 0.01 = 60000 Pa (1). (iv) A camel's wide feet spread its weight over a large area, so the pressure on the sand is small; a heel has a small area, so the pressure is large (2). | 1+1+1+2 |
| 20 | Cow: area 0.04 square metre (1); Cow pressure = 4000/0.04 = 100000 Pa (1); Man = 800/0.04 = 20000 Pa (1); Cow's pressure is five times (1); Cow may sink in wet soil more easily (1). | 1+1+1+1+1 |
| | **OR:** Equator heated more (1); Hot air rises (1); Cool polar air moves toward equator (1); Rotation of the Earth deflects (1); Global wind patterns result (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Writing P = A / F or forgetting that the area must be in square metres (convert cm squared first).
- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.

- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 0.02 square metre | 800 / 40000. |
| 2 | **(a)** less air lies above | Smaller air column. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | Energy supply cut. |
| 4 | **(a)** 1000000 Pa | 500/0.0005. |
| 5 | **(b)** storm surge | Low pressure and strong winds push water. |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 7 | **(b)** twice | Proportional to depth. |
| 8 | **(c)** anemometer | Measures wind speed. |
| 9 | **(d)** A is false but R is true. | A is false and R is true: liquids press on the walls as well as the base. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both true, but the fall of pressure with height does not explain why air exerts pressure. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Rotation of the Earth deflects winds (1); Spiralling inward gives rotation (1). | 1+1 |
| 12 | Pressure and oxygen are low (1); The body adapts slowly (1). | 1+1 |
| 13 | Hot air expands, becomes less dense (1); Cooler air flows in to replace it giving wind (1). | 1+1 |
| 14 | Outside pressure is lower (1); Air inside expands (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Air is pushed out leaving low pressure under the cup (1); Outside pressure holds it (1); Cracks let air in, so the seal fails (1). | 1+1+1 |
| 16 | Pressure change is large (1); Gases in the body expand (1); Body structure is damaged (1). | 1+1+1 |
| 17 | Snowshoes increase the area (1); Pressure on snow falls (1); Less sinking (1). | 1+1+1 |
| | **OR:** Pressure is proportional to depth (1); 9/3 = 3 times (1); Pressure = 3P (excluding air) (1). | 1+1+1 |
| 18 | Very low pressure inside (1); Fast rotating winds (1); Buildings explode and objects lift (1). | 1+1+1 |
| | **OR:** Atmospheric pressure pushes the card up (1); It exceeds the weight of the water (1); Seal prevents air entry (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Over warm oceans (1). (ii) The eye (1). (iii) The centre is a region of very low pressure and air moves from high to low pressure (1). (iv) It no longer receives warm moist air/water vapour from the sea, and friction with land slows the winds (any two) (2). | 1+1+1+2 |
| 20 | Collapsing can or inverted glass with card (1); Pressure outside exceeds inside (1); Straw or syringe (1); Suction cup (1); Application: pressure cooker or vacuum cleaner (1). | 1+1+1+1+1 |
| | **OR:** Pressure increases with depth (1); Thick strong walls (1); Pressure acts in all directions (1); Leak-proof seals (1); Check systems before dive (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Writing P = A / F or forgetting that the area must be in square metres (convert cm squared first).
- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.

- Confusing pressure (N/m squared, pascal) with force (N, newton).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** burst outwards | Pressure imbalance. |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 3 | **(a)** 2000 N | 200000 x 0.01. |
| 4 | **(a)** moist air rises and cools there | Orographic rain. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(d)** there is less air above the hill station | Air column is shorter. |
| 7 | **(d)** air pressure and oxygen are low | Thin air. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 9 | **(c)** A is true but R is false. | A is true; R is false: the ocean supplies the energy, and the supply is cut off over land. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | Movement of air from high to low pressure evens out the difference. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | The wind returns from the other side (1); People may leave shelters too early (1). | 1+1 |
| 12 | Difference = 55 kPa (1); F = 55000 x 0.1 = 5500 N (1). | 1+1 |
| 13 | Pressure is proportional to depth (1); It is four times (1). | 1+1 |
| 14 | Large pressure at the base (1); Curved thick wall resists the push (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Summer: land hot, low pressure; wind from sea to land (1); Brings moist air and rain (1); Winter: land cold; wind from land to sea (1). | 1+1+1 |
| 16 | Water pressure at depth is large (1); Thick walls withstand it (1); Crew safe (1). | 1+1+1 |
| 17 | Before: warm, humid, clouds build (1); During: lightning, rain, strong winds (1); After: cooler, clearer air (1). | 1+1+1 |
| | **OR:** Difference = 6 kPa (1); Air flows from high to low pressure (1); Strong inward winds (1). | 1+1+1 |
| 18 | 4 N per square cm (1); 12 N / 0.0003 square m (1); = 40000 Pa (1). | 1+1+1 |
| | **OR:** Charge may travel through wires (1); Risk of electric shock (1); Surges can damage appliances (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) The bottom-floor tap (1). (ii) Depth of the liquid (height of the liquid column above the point) (1). (iii) No, it presses on the base and walls in all directions (1). (iv) The bottom tap is deeper below the water surface in the tank, so the water pressure there is greater and pushes the water out more strongly (2). | 1+1+1+2 |
| 20 | Land heats quickly (1); Hot air rises, low pressure (1); Sea air flows inland (1); Night: land cools, sea warmer (1); Land breeze towards the sea (1). | 1+1+1+1+1 |
| | **OR:** Weather bulletins (1); Mock drills (1); Safe shelter identification (1); Emergency kit (1); Lightning safety rules (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Writing P = A / F or forgetting that the area must be in square metres (convert cm squared first).
- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.

- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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